



Sentinel

Web Security: Login

DEFENDING OUR DIGITAL WAY OF LIFE

Authentication

How do you login to your favourite application?



An Analogy: E-learning

You are required to submit your assignment on the e-learning portal. Before you are able to do so, you are required to identify yourself by entering your username and password.

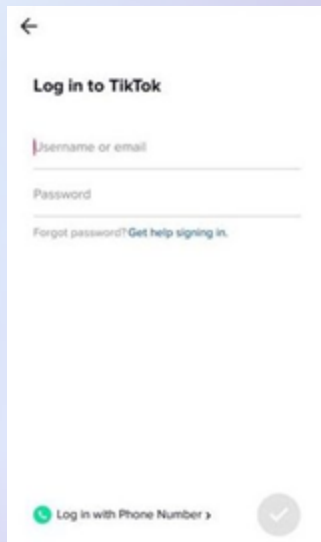
In Practice

- Client (browser): Browse to the e-learning login site and submit the credentials.
- Server: Perform authentication and return the results
- Client (browser): Has access to view the assignment page



Authentications

There are many ways of performing authentication, but the most popular way is by asking the user to log-in with a **password**.



←

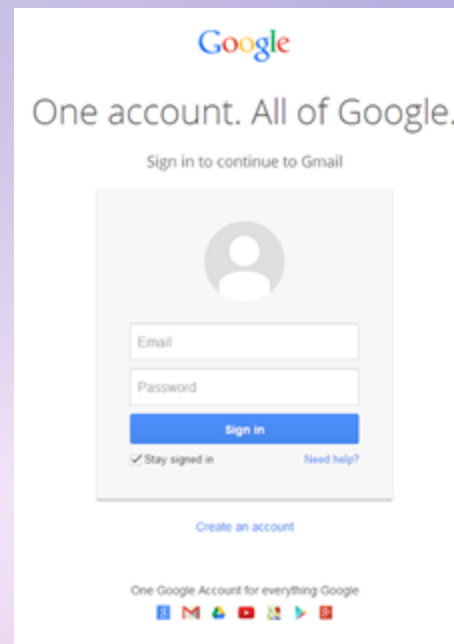
Log in to TikTok

Username or email

Password

[Forgot password? Get help signing in.](#)

Log in with Phone Number



Google

One account. All of Google.

Sign in to continue to Gmail

Sign in

☒ Stay signed in [Need help?](#)

[Create an account](#)

One Google Account for everything Google



Instagram

Username

Password

Log In

[Forgot your login details? Get help signing in.](#)

OR

Log in with Facebook

[Don't have an account? Sign up.](#)

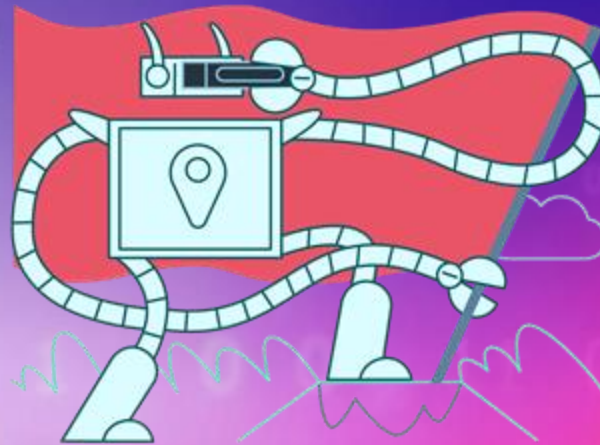
Keys To The Kingdom

Very simply put, passwords are like keys that allow us to open digital doors behind which information is hidden.

The password is a secret known only to the real person associated with the user.



Demo #1



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A Problem

In the previous demo, we saw that our username and password were hardcoded.

As a result, any changes made to the credentials will be lost when the server restarts. In other words, our data is **volatile**.

How can we solve this? Any ideas?

Data Persistency

In order to store our data in a way that preserves it, even if the server restarts (and even if it crashes), we can save our data on the hard disk.

In other words, we will save user data inside a file.

That way, the data of our users will be **persistent**.

An Analogy: Willie's Wonder Waffles

Your favourite restaurant: "Willies Wonder Waffles", wants to implement a membership program

Customers can visit the webpage to check their membership tier by entering their username and date of birth

Willies Wonder Waffles - Membership

Username :

Enter date of birth (DD/MM/YY)

submit

Where to store the information?

Answer: In a file!

The data file for the Willie Membership Program to will look as follows:



```
1 [
2   {
3     "username": "john",
4     "password": "password",
5     "dob": "13/01/1993",
6     "tier": "Classic"
7   },
8   {
9     "username": "april",
10    "password": "P@ssw0rd",
11    "dob": "28/06/2004",
12    "tier": "Premium"
13  }
14 ]
```

Looks familiar?

But wait! This looks familiar...

The contents of our file look just like a regular JavaScript object!



```
1 [
2   {
3     "username": "john",
4     "password": "password",
5     "dob": "13/01/1993",
6     "tier": "Classic"
7   },
8   {
9     "username": "april",
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11    "dob": "28/06/2004",
12    "tier": "Premium"
13  }
14 ]
```

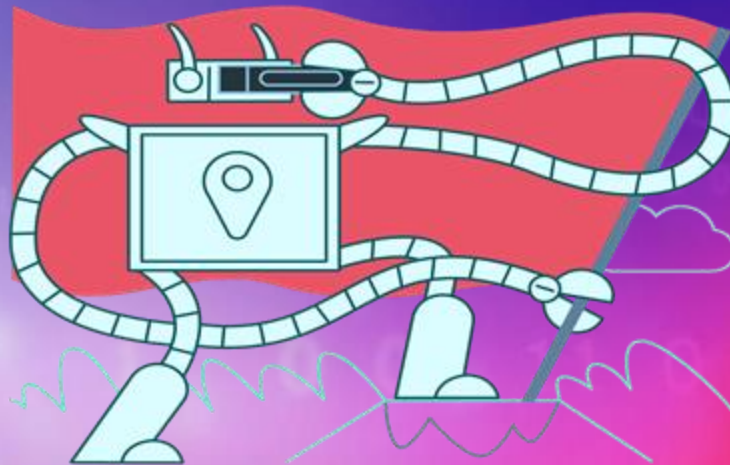

JSON

In JS, we can easily save objects into files by converting them into strings using the **JSON** module:

```
let data = [  
  { "username": "john", "dob": "...", ... },  
  { ... },  
  ...  
]  
JSON.stringify(data)
```

Then we can save the resulting string using the **fs** module, which we are already familiar with.

Demo #2



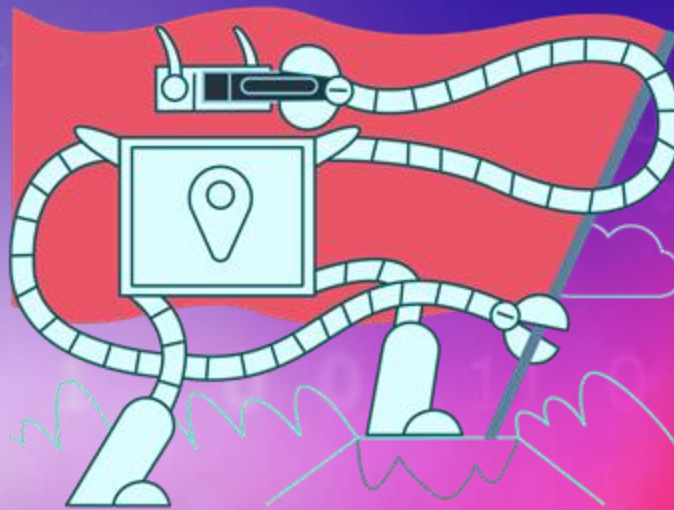
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Object into files: Example

And now let's save the data into a file!

```
const fs = require('fs')  
  
let data = [ ... ]  
let serialized = JSON.stringify(data)  
fs.writeFileSync('data.json', serialized)
```


Demo #3



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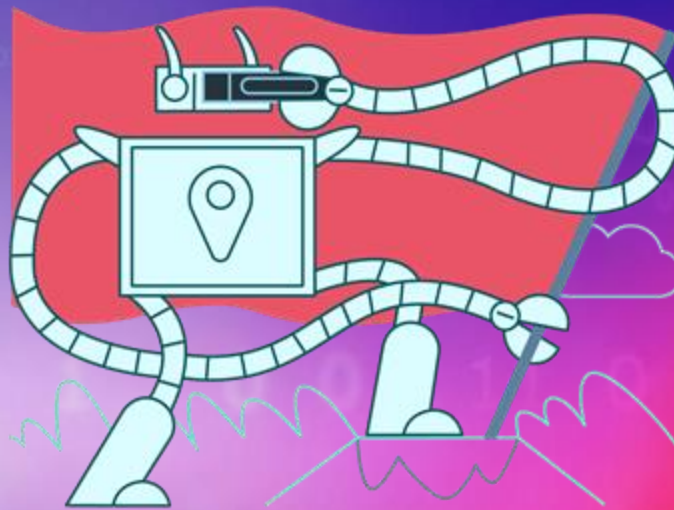
Reading JSON from a file

In order to use the data in our .json file, we need to read the file contents.

Then we need to convert the string we get back into a JS object. This is also easy:

```
const fs = require('fs')  
  
let fileContent = fs.readFileSync('data.json')  
let data = JSON.parse(fileContent)
```

Demo #4



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A Word on Security

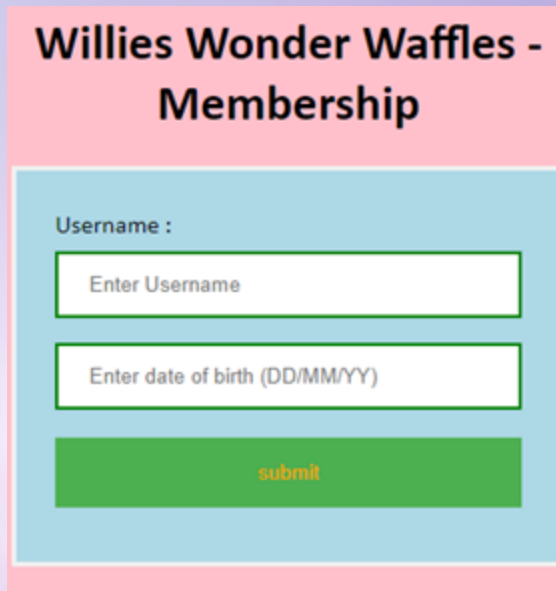
Our JSON files will contain sensitive information about our users.

Therefore we must not make them accessible outside the server.

Can we put our JSON in the static folder?

What's Next?

With that setup, when the user presses the submit button in Willie's website, the form will be submitted via POST request, containing the given username and date of birth.

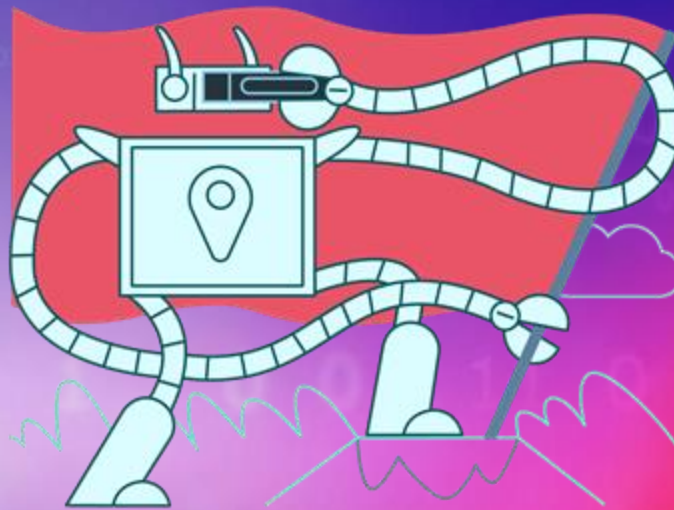


The image shows a web form titled "Willies Wonder Waffles - Membership". The form is contained within a light blue box. It has a pink header with the title. Below the title, there is a label "Username :" followed by a text input field with the placeholder "Enter Username". Below that is another text input field with the placeholder "Enter date of birth (DD/MM/YY)". At the bottom of the form is a green button with the text "submit" in yellow.

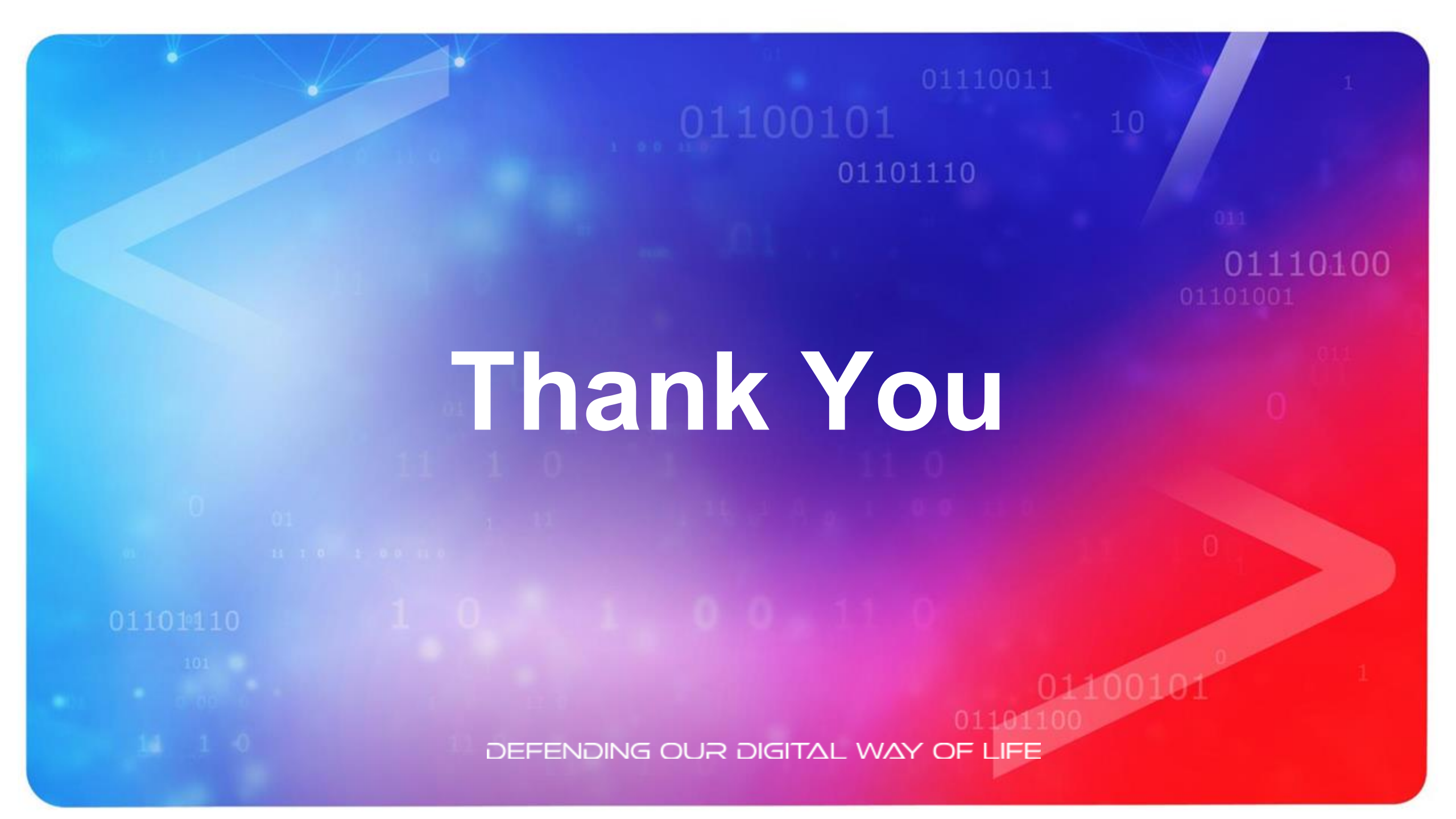
What's Next?

```
<form action="/check-membership" method="POST">
  <div class="container">
    <label>Username : </label>
    <input id="username" type="text" name="username"
placeholder="Username" required>
    <input id="dob" type="text" name="dob" placeholder="Date of
birth (DD/MM/YY)" required>
    <button type="submit">Submit</button>
  </div>
</form>
```

Demo #5



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Thank You

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